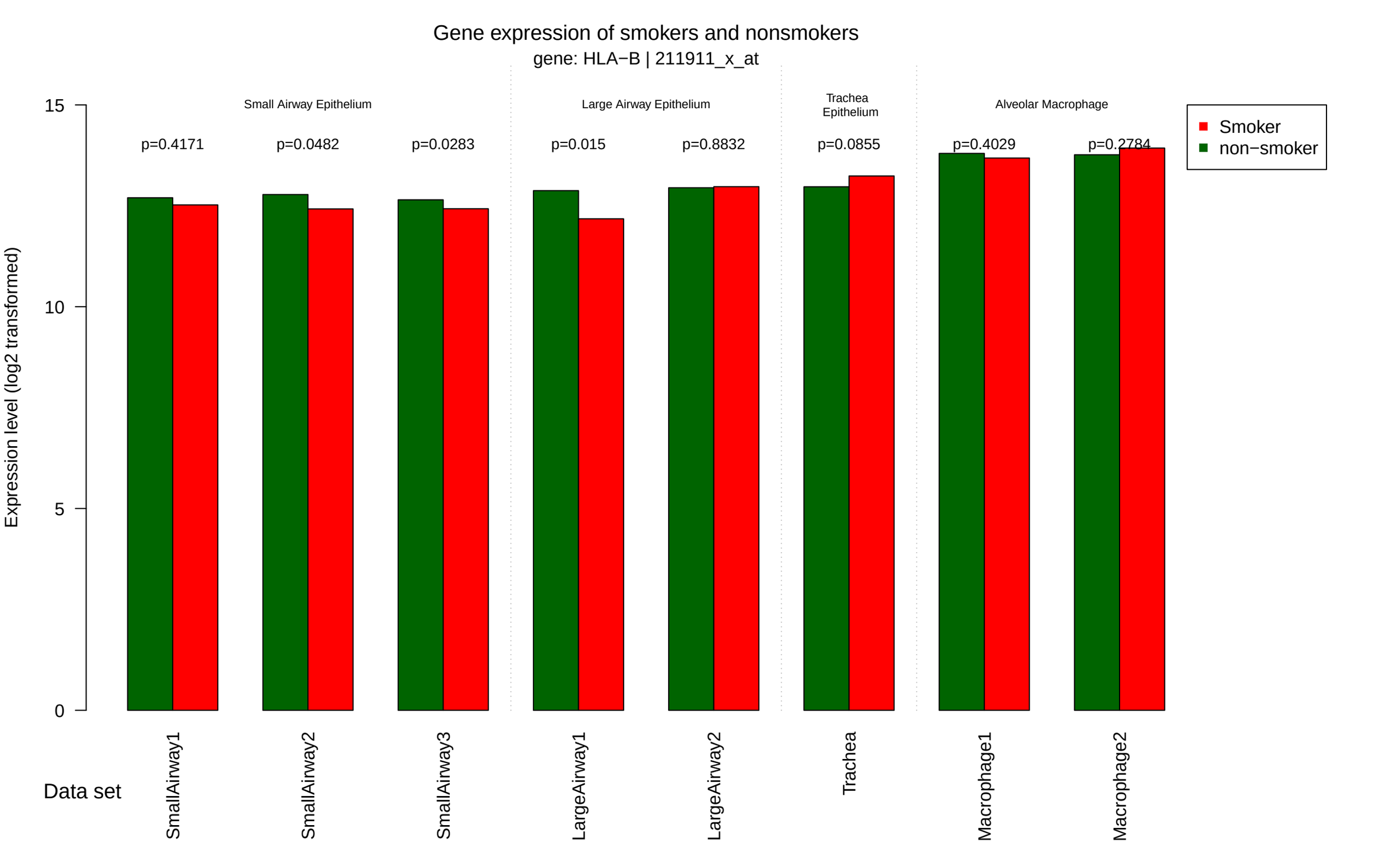
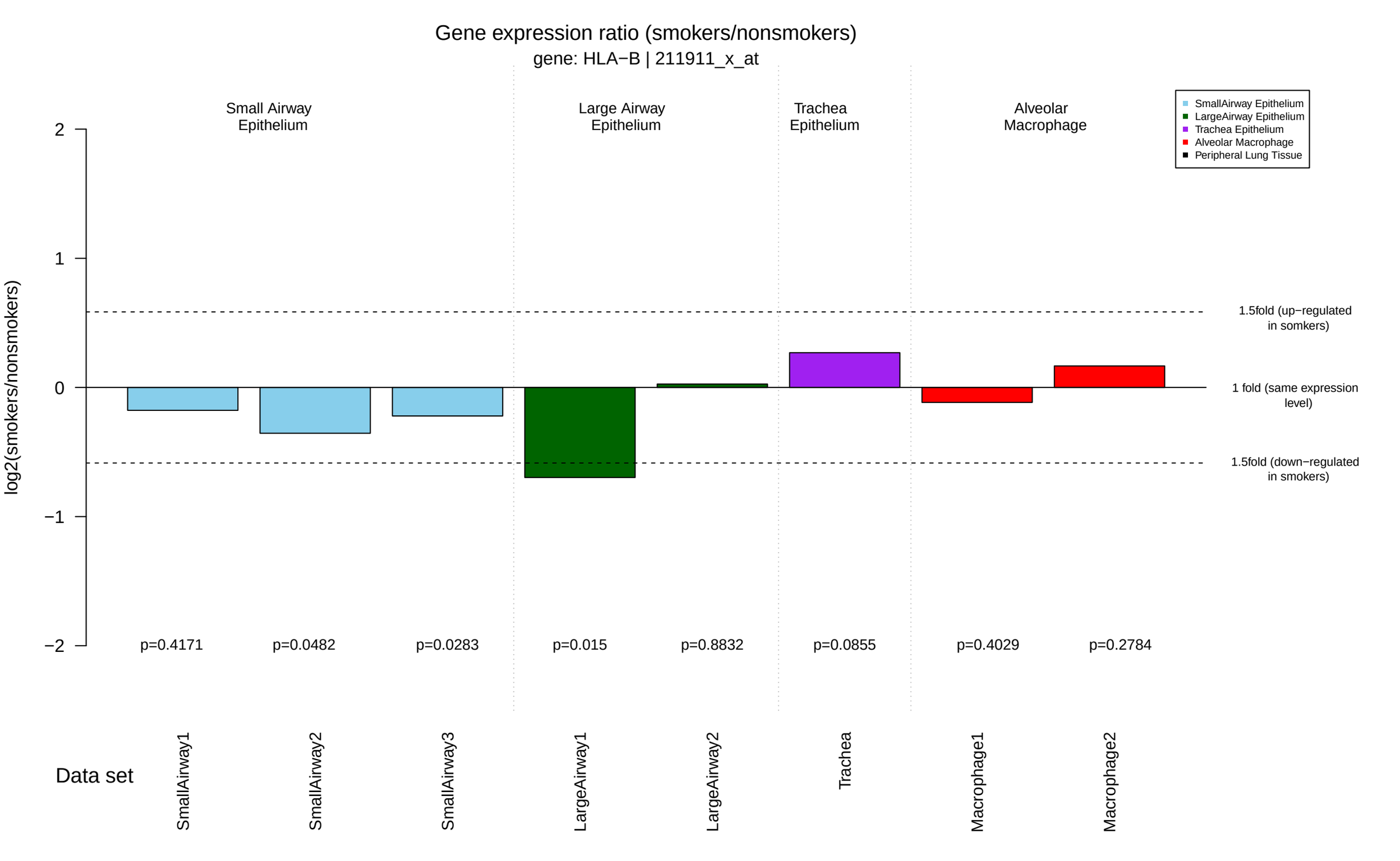
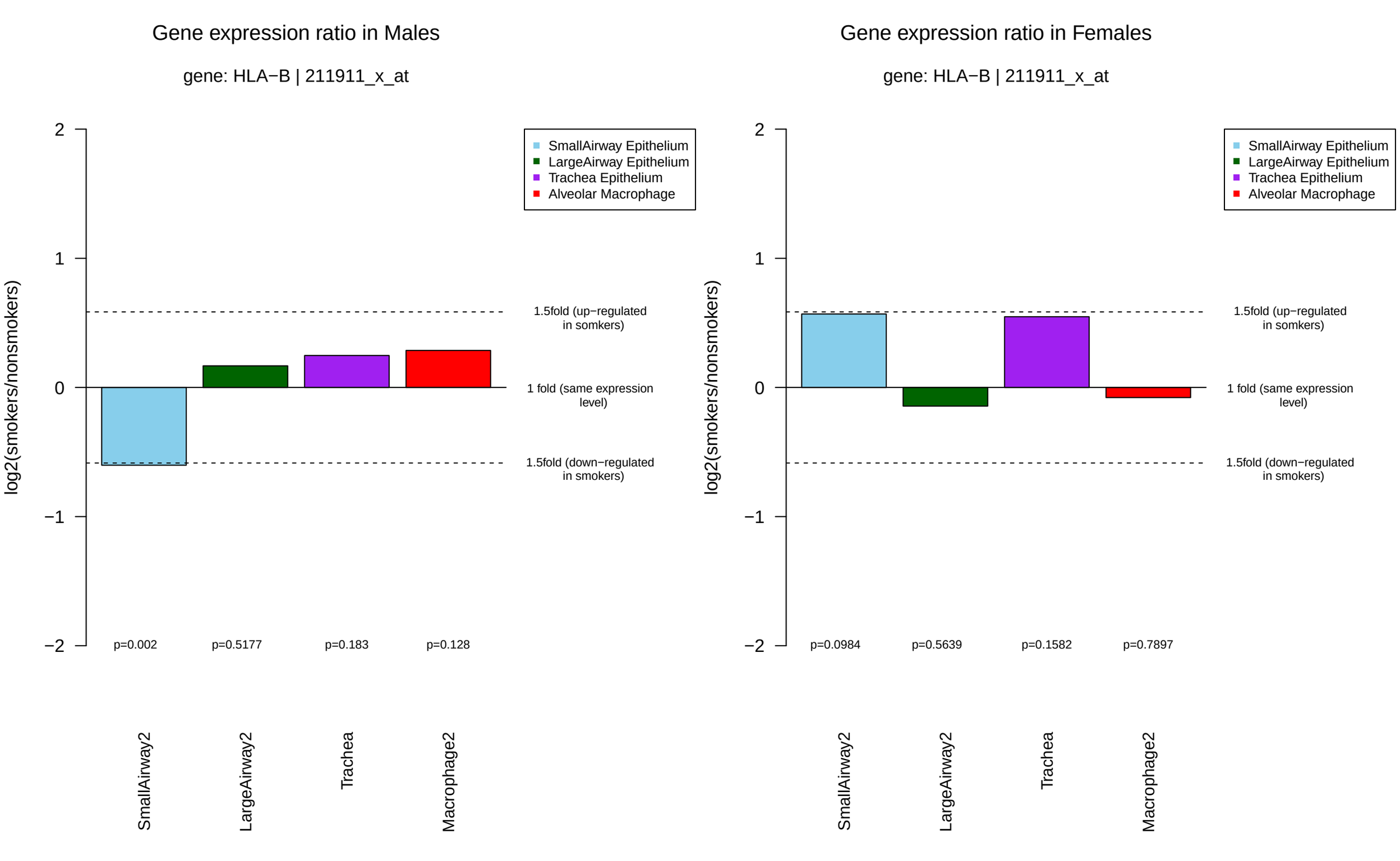
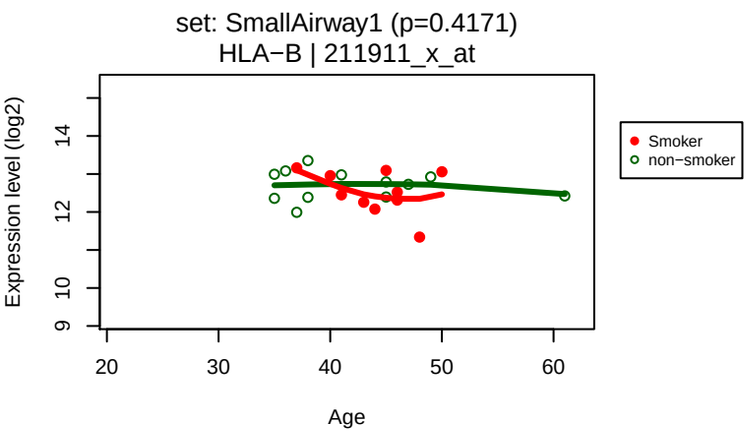
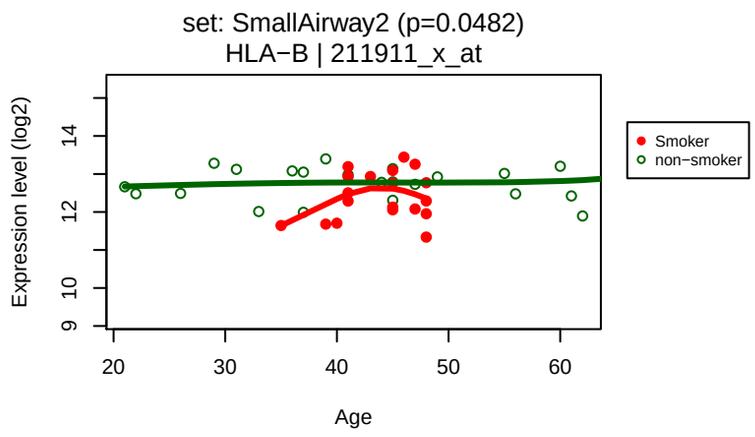
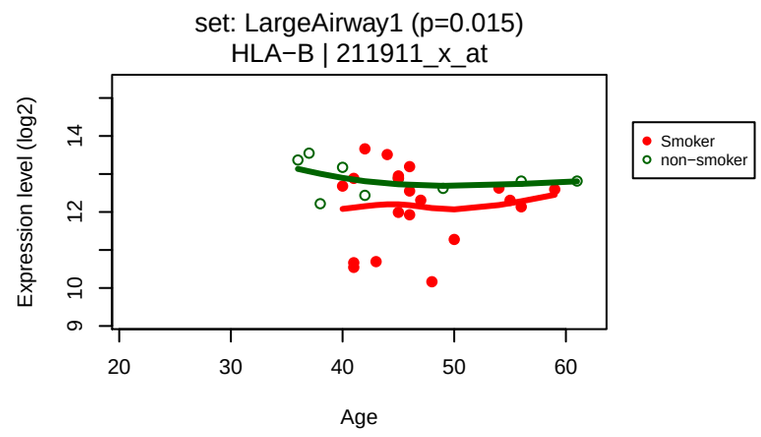

Gene expression of smokers and nonsmokers
gene: HLA−B | 211911_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4171
p=0.0482
p=0.0283
p=0.015
p=0.8832
p=0.0855
p=0.4029
p=0.2784
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

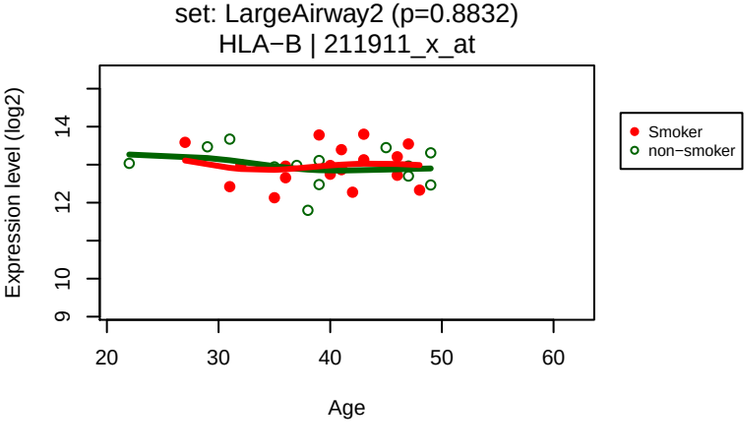

Gene expression ratio (smokers/nonsmokers)
gene: HLA−B | 211911_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4171
p=0.0482
p=0.0283
p=0.015
p=0.8832
p=0.0855
p=0.4029
p=0.2784
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

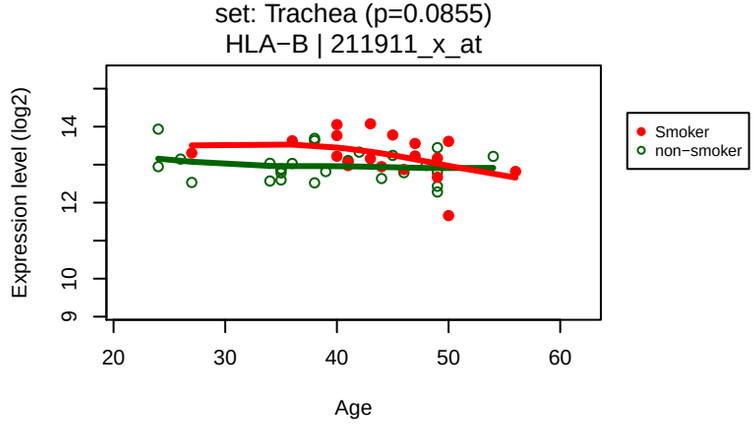

Gene expression ratio in Males
gene: HLA−B | 211911_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.002
p=0.5177
p=0.183
p=0.128
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−B | 211911_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0984
p=0.5639
p=0.1582
p=0.7897
SmallAirway2
LargeAirway2
Trachea
Macrophage2

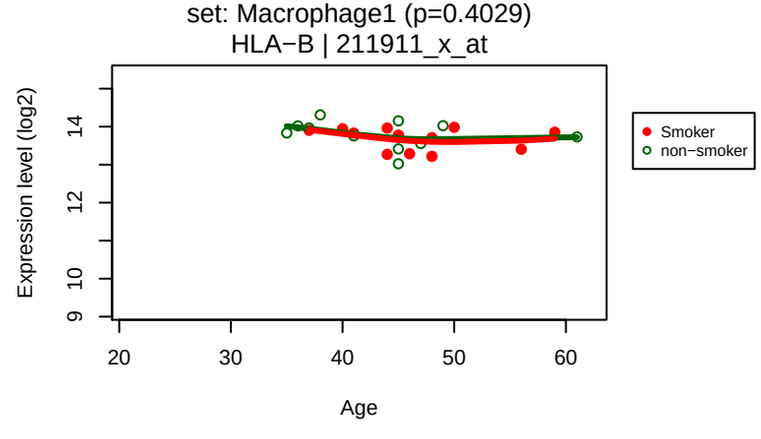

set: SmallAirway1 (p=0.4171)
HLA−B | 211911_x_at
Expression level (log2)
Age
Smoker
non−smoker

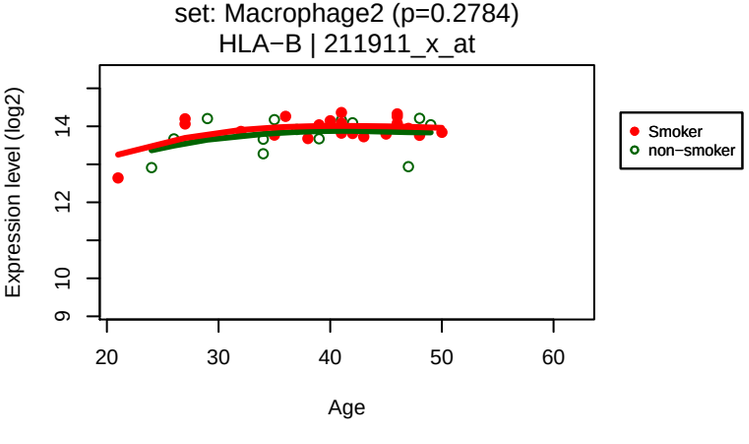

set: SmallAirway2 (p=0.0482)
HLA−B | 211911_x_at
Expression level (log2)
Age
Smoker
non−smoker

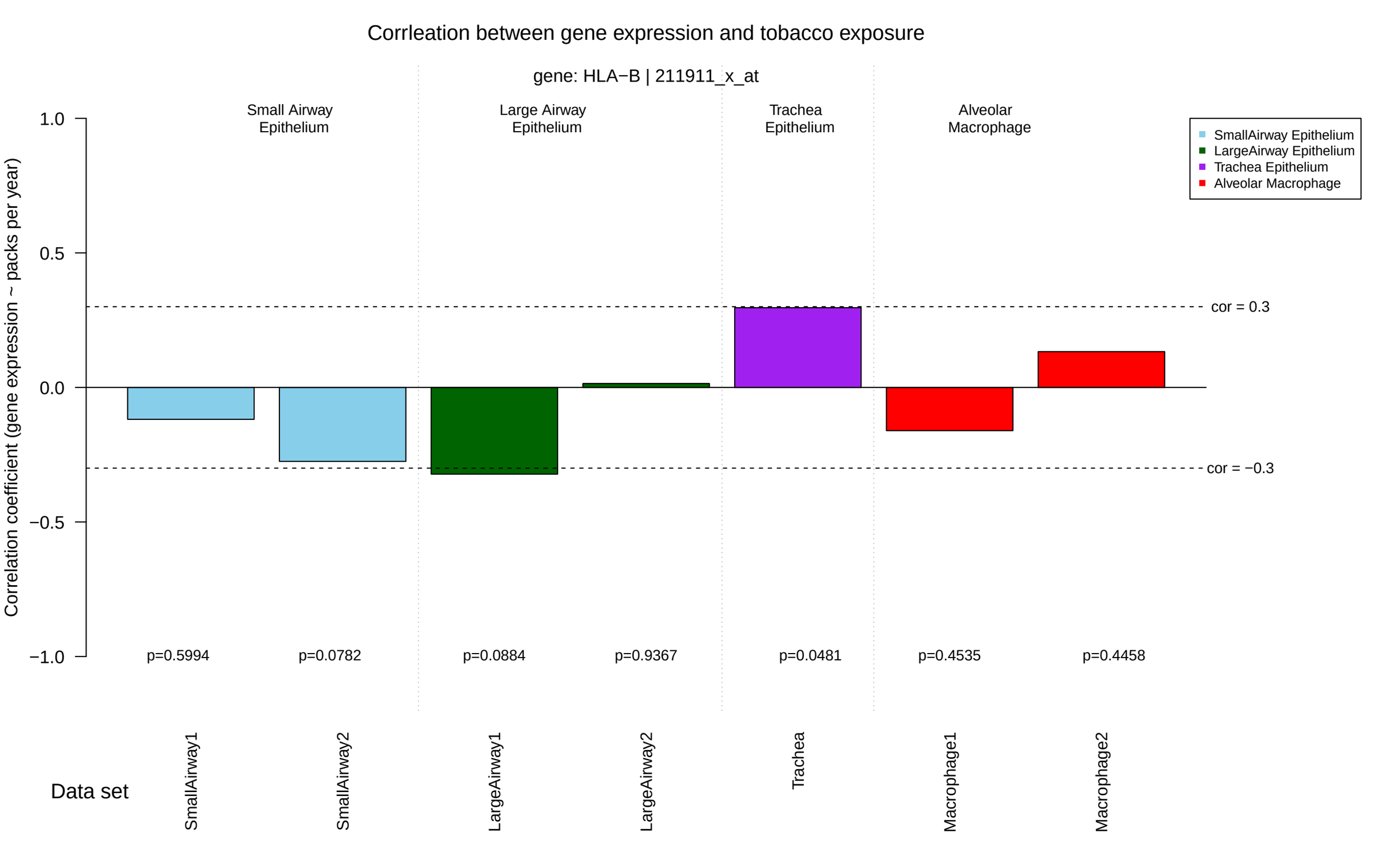

set: LargeAirway1 (p=0.015)
HLA−B | 211911_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8832)
HLA−B | 211911_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0855)
HLA−B | 211911_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4029)
HLA−B | 211911_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2784)
HLA−B | 211911_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−B | 211911_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5994
p=0.0782
p=0.0884
p=0.9367
p=0.0481
p=0.4535
p=0.4458
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2